

Raihan Fadhlurahman

Software Engineer

Bogor, Indonesia | 082240463327 | fadhlurahmanraihan29@gmail.com | linkedin.com/in/raihanfadhlurahman/

A results-driven Software Engineer with hands-on experience in backend development, system operations, and large-scale enterprise system environments. Proven track record in supporting mission-critical infrastructure, designing scalable system architectures, and translating complex business requirements into robust technical solutions.

Experienced in working within cross-functional teams across engineering, operations, and business stakeholders, with strong exposure to real-time data systems, distributed architecture (Kafka), and database management. Demonstrates strong analytical thinking, ownership, and adaptability in high-demand environments.

Passionate about building reliable, scalable, and efficient systems while continuously driving operational excellence and product improvement.

AREA OF EXPERTISE

- Backend Engineering & API Development
- System & Infrastructure Monitoring (HES, Kafka, VM, SCADA)
- Database Design, Optimization & Query Analysis
- Business & System Analysis (BRD, FSD, ERD, SDLC)
- Distributed Systems & Real-time Data Processing
- Automation & Workflow Orchestration (n8n, Scheduler)
- Software Development Lifecycle (Agile / Scrum)

EDUCATION

Politeknik Negeri Jakarta

Agustus 2020 - Agustus 2024

Bachelor's degree in Information Technology

- Cumulative GPA: 3.57/4.00
- Successfully obtained a bachelor's degree in 2024 by creating E-Ticketing application for application service requests provided by the Communication and Informatics Agency of Bogor Regency for agencies in the scope of work. There is an E-Ticketing feature with priority scale and email notification. There is a forum feature to exchange information and interact between agencies using the nested recursion method.
- Related coursework: Information Technology, Fullstack Developer, Database Administration, Web Application, Back-end Developer.

CCIT Fakultas Teknik Universitas Indonesia

Agustus 2020 - Agustus 2022

Associate's degree in Software Engineering

- Cumulative GPA: 3.56/4.00
- Certified as Software Engineering in Java Technologies.
- Certified as Professional Information Technology.

WORK EXPERIENCE

IT Operation & Maintenance • PT PLN Icon Plus

April 2025 - Now

Head End System (HES) Nari3000 – AMI Smart Meter Operation Center (SMOC)

- Conducted hourly operational monitoring and reporting for the Nari3000 Head End System (HES) within the AMI Smart Meter Operation Center (SMOC), ensuring continuous system availability and performance.
- Performed database inspections by reviewing logs, transaction flows, storage consistency, and overall system health to maintain data integrity across AMI infrastructure.
- Executed Kafka pipeline monitoring (consumer performance, idFront services, directory processes) to ensure real-time meter and DCU data ingestion remained stable and uninterrupted.
- Managed HES UI-based inspections, validating smart meter and DCU telemetry updates received every 15

minutes to ensure operational standardization and compliance.

- Monitored key operational KPIs, including:
 1. DCU Online Ratio maintained at $\geq 98\%$
 2. DCU Clock Synchronization achieving 100% accuracy (<10 seconds deviation)
 3. Continuous acceptance of scheduled meter data transactions
- Conducted Virtual Machine (VM) inspections, assessing CPU load, memory utilization, disk capacity, and system resource trends to prevent infrastructure bottlenecks.
- Performed field-to-system data synchronization checks using ACMT, ensuring alignment between AMI on-site meter readings and centralized system records.
- Supported troubleshooting and operational requests through SQL query execution, including targeted data extraction, anomaly investigation, and reporting support for stakeholders.
- Identified anomalies, escalated incidents promptly, and contributed to preventive maintenance actions to sustain reliability in a large-scale AMI ecosystem.

Project: PLN Edu+ Learning Management System (Business Analyst)

- Drove end-to-end business analysis for a full-scale Learning Management System (LMS), translating complex product vision into scalable, production-ready system architecture that aligns business objectives with technical execution.
- Acted as a key bridge between stakeholders and engineering teams, eliminating ambiguity in requirements and accelerating development cycles through clear, structured, and highly detailed system documentation.
- Owned the design of critical LMS ecosystems, including:
 - Course & Bundle Management (modular and scalable structure)
 - Advanced Quiz Engine (multi-attempt logic, randomized questions, strict validation scoring)
 - Assignment System (dynamic deadline, revision lifecycle, grading automation)
 - Student Progress Tracking (time-based validation, completion enforcement)
 - Certificate Engine (rule-based eligibility with grading classification)
 - Integrated Reporting & Ticketing System (chat-based support with automation & scheduler)
- Engineered complex business logic and system behaviors, such as:
 - Learning rollback mechanism (forcing content re-consumption upon quiz failure)
 - Dynamic assignment deadline calculation based on user start time
 - Bundle-course dependency handling for flexible product offerings
 - Automated ticket lifecycle (auto-response, auto-close via scheduler)
 - All-or-nothing scoring logic for multi-answer quiz validation
- Produced high-quality documentation including BRD, FSD, ERD, Activity Diagrams, and Business Process Flows, significantly improving requirement clarity and reducing miscommunication across teams.
- Designed scalable relational data models using ERD to support high-complexity features such as transactions, enrollments, bundles, reporting systems, and real-time progress tracking.
- Utilized Mermaid, Draw.io, and Lucidchart to create structured and developer-friendly diagrams, enabling faster understanding of system flows and reducing onboarding time for engineers.
- Introduced system-level innovations that improved efficiency and scalability, including:
 - Simplified checkout architecture (direct purchase without cart complexity)
 - Unified transaction and payment handling
 - Centralized progress tracking system across all learning modules
 - Modular feature design enabling future extensibility

- Collaborated cross-functionally with Frontend, Backend, and UI/UX teams, ensuring seamless API integration, consistent data flow, and intuitive user experience across the platform.
- Contributed to UX optimization using Figma, focusing on user flow clarity, usability, and reducing friction in key user journeys such as course consumption and assessment flows.
- Performed API validation and integration testing using Postman, ensuring data accuracy, system reliability, and smooth interaction between services.
- Managed and tracked development progress using Jira (Agile/Scrum), ensuring structured sprint execution and timely feature delivery.
- Demonstrated strong product thinking by not only translating requirements but also proactively identifying gaps, proposing solutions, and driving continuous improvement across the system.

Project: Automation & Real-Time Notification Initiative (SCADA Event Monitoring)

- Initiated the development of an automation-based alerting workflow using n8n, integrating operational SCADA event data with real-time executive notification.
- Leveraged the SCADA data stream, where primary outage/event indicators are first published through Kafka, to detect potential power outage events as early as possible.
- Designed an automated pipeline to extract and process SCADA outage timestamps, enabling near real-time insights into upcoming disruption events.
- Implemented a WhatsApp blasting notification system to proactively inform UP3 and ULP leadership stakeholders when outages are detected in their respective service areas.
- Enhanced response readiness and situational awareness by reducing manual reporting delays and improving real-time operational transparency.

Project: SynerGCG (Business Analyst & Technical Engineer) – PLN EPI (Energi Primer Indonesia)

- Served as a System Engineer under PT ICON+, responsible for analyzing existing business processes and ensuring alignment with the Business Requirement Document (BRD) as part of a large-scale system reengineering initiative.
- Actively contributed to the application development lifecycle using the Software/System Development Life Cycle (SDLC/SSDLC) methodology, supporting both planning and implementation phases.
- Developed and maintained key project documentation, including:
BRD (Business Requirement Document), FSD (Functional Specification Document), DFD (Data Flow Diagram), Code Review Reports, and other technical planning documents to ensure structured system design and compliance with project standards.
- Designed and enhanced UI/UX prototypes using Figma, improving workflow efficiency, usability, and user experience across the platform.
- Acted as a Copywriter and Documentation Specialist, producing detailed system requirement documentation, technical deliverables, and supporting materials for stakeholders and project governance.
- Gained hands-on experience as a Quality Assurance (QA) Engineer by conducting functional and regression testing to ensure system reliability and performance.
- Performed API testing and validation using Postman, ensuring accurate integration between backend services and client applications.
- Automated test execution and reporting using Katalon Studio, improving testing efficiency and supporting continuous quality improvement throughout the development process.

Project: Gotig (Business System Analyst Hybrid) – PLN EPI (Energi Primer Indonesia)

- Worked as a System Engineer / Project Coordination Lead under PT ICON+, responsible for defining the overall direction and functional scope of a new reengineered internal application through structured business planning and requirement gathering.
- Led the preparation of the Business Requirement Document (BRD), translating operational needs into clear system objectives, process flows, and functional priorities.
- Initiated and contributed to the development of Gotig, an internal platform supporting meeting attendance, daily activity reporting, support tracking, and operational transparency dashboards (MHCA) for comprehensive monitoring.
- Played a key role in shaping the system's business process design (Probis), ensuring that workflows, module integration, and user roles aligned with organizational goals.
- Designed UI/UX prototypes using Figma, providing consistent and user-friendly interfaces across all application modules.
- Acted as the main Communicator and Stakeholder Liaison between PLN EPI and partner companies, ensuring alignment of business expectations, technical understanding, and project deliverables.
- Supported documentation and deliverable preparation as a Copywriter, producing structured system descriptions and supporting materials for internal and external stakeholders.
- Performed Quality Assurance (QA) activities, including functional and regression testing, to ensure application stability and compliance with defined requirements.
- Conducted API testing and integration validation using Postman, ensuring reliable communication between system modules and backend services.
- Automated testing workflows and generated execution reports using Katalon Studio, improving testing efficiency and supporting continuous quality improvement throughout the project lifecycle.

Web Developer • Freelance

Januari 2025 - Februari 2025

- Developed a company profile website for a private land management company using Laravel and Tailwind, with MySQL database management. Built a CMS-based website to streamline project management and file storage for users.

Web Developer • PT Summit Citra Digital

Januari 2025 - Februari 2025

- Conducted slicing for the Setwan District Council website in Bogor using HTML and Tailwind as the initial reference for the development of a Content Management System (CMS).
- Developed a Content Management System (CMS) using Laravel and MySQL database as a subsequent phase after slicing. This system is designed to manage the content to be displayed to the public by the Setwan District Council in Bogor.

Backend Engineer • Diskominfo Kabupaten Bogor

Agustus 2023 - Juli 2024

- Developed the backend for the E-Office application, managed by Dinas Komunikasi dan Informatika Kabupaten Bogor (available on both web and mobile platforms) using Golang (Google Language) and MySQL database.
- Designed and structured the database architecture to ensure optimal data relationships and seamless integration across multiple systems.
- Developed a backend system to bridge communication between mobile and web applications, ensuring seamless data exchange and synchronization. Optimized the system to handle high traffic during peak transactions, reducing load times and enhancing user experience.
- The application serves approximately 150 daily users for tasks such as attendance, assignment tracking, activity logging, and employee evaluations.
- Installed and configured Ubuntu servers for database storage and application deployment, ensuring smooth operation and robust performance in a production environment.
- Implemented cron jobs on the server to schedule regular database backups twice a month (on the second and fourth

weeks) to safeguard critical data.

Backend Engineer • Istana Thrift

Maret 2024 – April 2024

- Developed the backend for a Point of Sales (POS) system using the Laravel framework for the cashier section of Istana Thrift store.
- Designed an efficient database structure to manage member data and sales transactions for the store.
- Created business logic workflows within the application to handle payment processing, including member discounts, product discounts, pricing, and payment methods.

Fullstack Engineer • Golongan Karya (Golkar)

Januari 2024 - Februari 2024

- Developed the GARPPU (Gerakan Pemantau Pemilu) application for Golkar Party's internal use, managing volunteers at each polling station for DPR and DPRD elections, utilizing the Laravel framework. The application supports approximately 8 users.
- Designed an efficient database to store volunteer and voter data, as well as election result proofs (C1 documents) from each polling station.
- Created a real-time election results page to display provisional results as they are reported from various polling stations.
- Created a user-friendly interface with an intuitive layout, ensuring that features were placed in a logical flow for easy submission of result proofs.

Backend Engineer • PT Nusa Tekno Global

Mei 2023 - Juli 2023

- Developed the backend for the E-Event application using Golang (Google Language), providing a platform where users can view upcoming events.
- Designed an efficient PostgreSQL database to store event data, organizers, and event participants.
- Created Restful APIs to integrate data seamlessly with the frontend for smooth user interaction and data retrieval.

ORGANIZATIONAL EXPERIENCE

Himpunan Mahasiswa Teknik Informatika

Agustus 2022 - Agustus 2023

Sosial and Political Staff

- Organized an event and became responsible for enlivening Pancasila Day with more than 15 participants.
- Socializing political issues that occur in Indonesia to provide important information to all Informatics Engineering students.
- Volunteering as operational staff at TIK Games 2022 event.

Sahabat PNJ (SPNJ)

Android Studio - Member

Agustus 2021 - Mei 2022

- Members of Android Developer by learning the basics of creating mobile-based applications using Android Studio and creating small-scale application projects, namely the creation of building area calculation applications.

Technical Skills

Backend & API Development

Golang, Java, PHP (Laravel), RESTful API Design, Microservices Architecture

Database & Data Management

MySQL, PostgreSQL, Query Optimization, Database Design (ERD), Transaction Handling

System & Infrastructure

Linux (Ubuntu Server), VM Monitoring, System Performance Analysis, Server Management

Distributed Systems & Streaming

Apache Kafka (Monitoring & Data Pipeline), Real-time Data Processing

Automation & Tools

n8n (Workflow Automation), Cron Jobs, Postman (API Testing), Katalon (Automation Testing)

Software Engineering Practices

Object-Oriented Programming (OOP), Clean Code, SDLC, Agile/Scrum

Languages

- Indonesia : Native
- English : Limited working proficiency (TOEFL Score 503)

Certifications

- DNIIT Software Engineering – Java Technologies | CCIT-NIIT
- Web Developer | Badan Nasional Sertifikasi Profesi
- Pemrograman HTML 5 | Udemy
- Object Oriented Programming | Udemy